

A Comprehensive Review of Sensor Technologies for Wastewater Collection Networks: Operating Principles, Field Performance, and Failure Dynamics

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Abstract

Real-time monitoring of underground wastewater collection networks (WCNs) is essential for mitigating sanitary sewer overflows (SSOs), detecting early pipe chokes, managing hydraulic capacity, tracking industrial contaminant discharges, and controlling biogenic sulfuric acid corrosion (H_2S). However, raw municipal sewage represents an exceptionally hostile, biologically active, and geometrically constrained environment that severely challenges sensory instrumentation. This paper provides a comprehensive, evidence-based literature review of the full spectrum of sensing technologies deployed across urban wastewater networks. We systematically examine the operational physics, transducer mechanics, laboratory validation, and field-trial performance of continuous level sensors (non-contact 77–81 GHz FMCW radar, non-contact ultrasonic transit-time, submersible hydrostatic pressure, and optical time-of-flight) in direct contrast with discrete binary pulse/float devices (mechanical float switches, mercury-free tilt switches, and conductivity pulse electrodes). In addition, we review open-channel velocity and discharge instrumentation (submerged area-velocity Doppler probes, non-contact surface radar, and the physical breakdown of level-derived Manning approximations) alongside non-invasive pipe condition and blockage screening tools (acoustic pulse reflectometry, ultrasonic guided waves, CCTV computer vision, and distributed fiber-optic DTS/DAS). Furthermore, we analyze online water quality probes (toroidal conductivity, glass-bulb pH/ORP, optical luminescent DO), advanced optical spectroscopy (UV-Vis absorbance and 2D-fluorescence EEM spectroscopy for instantaneous BOD_5 surrogacy), microbial fuel cell (MFC) biosensors, and gas/liquid hydrogen sulfide monitors. Environmental failure modes—including thermal/humidity gradient refraction, condensation blinding, foam absorption, near-field dead bands, fats, oils, and grease (FOG) encrustation, ragging, and sensor drift—are critically reviewed together with subterranean wireless telemetry (LoRaWAN/NB-IoT) attenuation and battery energy budgeting.

Keywords: Wastewater sensors; Continuous level radar; Ultrasonic level; Float and tilt switches; Area-velocity flow; Optical spectroscopy; 2D-Fluorescence; Hydrogen sulfide sensing; Biofouling and ragging.

1 Introduction and Operational Sewer Environment

Wastewater collection networks (WCNs) form the subterranean lifeline of modern urban sanitation. Comprising thousands of kilometers of interconnected gravity sewers, drop manholes, siphon chambers, and pressurized pumping mains, these systems operate under extreme environmental, biological, and chemical stress [14, 18]. Sewer conduits convey complex three-phase matrices consisting of turbulent liquid effluents, suspended organic and inorganic grit, floating fats, oils, and grease (FOG), and aggressive headspace gas mixtures laden with moisture, hydrogen sulfide (H_2S), and volatile organic compounds [1, 8].

When hydraulic conveyance fails due to pipe chokes, FOG blockages, root intrusions, structural collapses, or severe stormwater infiltration and inflow (I&I), the result is catastrophic sanitary sewer overflows (SSOs), surface flooding, and environmental contamination [1]. Historically, water utilities managed sewer networks through reactive customer complaint dispatches and scheduled manual inspections. Over the past decade, advancements in low-power microelectronics, high-frequency electromagnetic radar, piezoelectric acoustic transducers, optical spectrometers, and low-power wide-area network (LPWAN) telemetry have enabled the transition toward continuous digital sewer monitoring [6, 8].

However, deploying sensors inside municipal sewers presents formidable physical hurdles:

- **Aggressive Atmospheric Chemistry:** Continuous relative humidity (95%–100% RH) combined with biogenic sulfuric acid creates severe condensation and corrosive attack on sensor housings and electronic connections [8].

- **Submerged Matrix Hazards:** Submerged probes experience rapid biological biofilm formation within 24–48 hours, FOG deposition, and physical ragging (the entanglement of non-flushable fibrous wipes around sensor cables and brackets) [1, 14].
- **Subterranean Communication Barriers:** Thick reinforced concrete and cast-iron manhole covers attenuate wireless radio signals by 20–40 dB, restricting telemetry bandwidth and battery lifespans [12].
- **Confined Space Access and Safety:** Entry into sewer chambers requires specialized safety equipment, traffic management, and hazardous atmosphere (ATEX / IECEx Zone 0/1) explosion-proof certification, making frequent manual maintenance economically prohibitive [18].

This paper synthesizes the empirical evidence, transducer physics, and field trial findings across a curated corpus of 23 authoritative, peer-reviewed cornerstone research studies. To eliminate analytical redundancy and maintain a strict focus on physical sensory hardware, all mathematical placement algorithms and optimization models are decoupled from this review. Instead, we examine the concrete findings from real-world municipal field trials (including the Shoalhaven Water radar network, Hong Kong contactless flow campaigns, and the UWO urban hydrology field laboratory), controlled transducer characterizations, and long-term failure evaluations in subterranean sewer environments.

2 Continuous Level and Depth Sensing Technologies

Continuous depth measurement provides the foundational time series $h(t)$ required to monitor diurnal dry-weather flow hydrographs, detect hydraulic surcharge, track rate-of-rise ($\frac{dh}{dt}$), and identify upstream flow restrictions [3, 8]. Four primary continuous depth measurement modalities exist in wastewater engineering.

2.1 Frequency-Modulated Continuous-Wave (FMCW) Radar

Non-contact FMCW radar has emerged as the premier reference technology for sewer depth monitoring. Operating in microwave frequency bands (typically 77–81 GHz, with older systems at 24 GHz or 60 GHz), the radar transmits a continuous electromagnetic chirp whose frequency sweeps linearly over bandwidth B across sweep duration T_{sweep} [8]. The electromagnetic wave reflects off the wastewater surface dielectric boundary and is mixed with the instantaneous transmitted signal to generate an intermediate beat frequency f_b :

$$d = \frac{c \cdot f_b \cdot T_{\text{sweep}}}{2 \cdot B} \quad (1)$$

where $c \approx 3 \times 10^8$ m/s is the speed of light in air. Flow depth is determined by subtracting measured distance d from the known manhole invert elevation H_{invert} : $h(t) = H_{\text{invert}} - d(t)$.

Empirical Field Findings and Advantages:

- *Narrow Beam Directivity:* Operating at 77–81 GHz (millimeter-wave band) allows compact antenna lens arrays to achieve exceptionally narrow beam divergence angles (3°–8°) [8]. In Australian utility field trials conducted by Drenoyanis et al. [8] across the Shoalhaven Water network, 80 GHz radar successfully isolated narrow sewer inverts in deep, obstructed manholes without capturing false echoes from ladder rungs, step irons, or lateral benching walls.
- *Atmospheric Immunity:* Electromagnetic microwave propagation is unaffected by air temperature fluctuations, steep vertical thermal gradients, dense steam, pressure differentials, or toxic H_2S /methane gases.
- *Surface Dielectric Reflection:* Water has a high relative permittivity ($\epsilon_r \approx 80$), providing a sharp reflection boundary even when covered with thin grease or foam layers.

Operational Limitations and Energy Demands: High-frequency radar digital signal processors (DSP) require peak currents of 15–35 mA during active signal sweeps (50–100 mW). To achieve multi-year battery autonomy on primary $Li - SOC l_2$ battery packs, radar nodes operate in pulsed duty-cycled modes (e.g., waking every 2 to 5 minutes for a 500 ms measurement cycle). Capital purchase costs remain higher than acoustic or hydrostatic alternatives.

2.2 Non-Contact Ultrasonic Transit-Time Level Transducers

Ultrasonic level sensors emit high-frequency acoustic wave bursts (typically 20–200 kHz) from a piezoelectric transducer mounted at the manhole roof toward the water surface. Distance is derived from round-trip time-of-flight (ToF) Δt :

$$d = \frac{c_{\text{air}}(T) \cdot \Delta t}{2} \quad (2)$$

where acoustic velocity in air c_{air} is governed by thermodynamic temperature T (Kelvin):

$$c_{\text{air}}(T) = \sqrt{\frac{\gamma RT}{M}} \approx 331.3 \sqrt{1 + \frac{T_c}{273.15}} \quad (\text{m/s}) \quad (3)$$

where T_c is temperature in $^{\circ}\text{C}$, $\gamma \approx 1.4$ is the heat capacity ratio, R is the universal gas constant, and M is the molar mass of air.

Documented Operational Failure Modes in Sewers:

- *Thermal and Humidity Stratification Errors:* Enclosed manholes exhibit steep vertical temperature gradients between the warm sewage base ($\approx 18\text{--}22^{\circ}\text{C}$) and the cold road surface cover ($\approx 0\text{--}10^{\circ}\text{C}$). Because sound velocity varies continuously along the acoustic path, a single temperature compensation thermistor located on the transducer face produces systematic distance errors between 5% and 12% [1, 8].
- *Condensation and Transducer Blinding:* Saturated relative humidity (100% RH) causes liquid water droplets to condense directly across the piezoelectric diaphragm face. This moisture layer dampens mechanical oscillation, causing ringing noise that extends the near-field “dead band” (the blind zone immediately below the sensor) from 0.15 m to > 0.5 m, blinding the sensor during critical high-water surcharge events [23].
- *Acoustic Absorption by Biological Foam:* Organic surfactants and biological foaming absorb high-frequency acoustic compression waves, causing signal dissipation and intermittent false zero returns.
- *Beam Spreading False Echoes:* Typical ultrasonic beam angles ($10^{\circ}\text{--}15^{\circ}$) create wide acoustic cones that reflect off manhole walls, benching ledges, and debris, requiring complex echo-profile filtering algorithms.

2.3 Submersible Hydrostatic Pressure Transducers

Hydrostatic level probes utilize a piezoresistive silicon strain gauge or ceramic capacitive diaphragm submerged at the pipe invert. The probe measures total hydrostatic pressure P_{total} exerted by the water column h :

$$h = \frac{P_{\text{total}} - P_{\text{atm}}}{\rho(T, S) \cdot g} \quad (4)$$

where ρ is sewage fluid density (a function of temperature T and suspended solids S), g is gravitational acceleration, and P_{atm} is atmospheric surface pressure supplied via a vented internal capillary tube in the signal cable.

Field Evaluation and Maintenance Burden: Hydrostatic probes draw minimal electrical power (< 15 mW) and are immune to headspace vapor and foam. However, long-term sewer trials reveal severe operational vulnerabilities [8, 14]:

1. *FOG and Silt Encrustation:* Fatty grease and biological slimes coat the flexible sensing diaphragm, dampening pressure transmission and causing positive offset drift.
2. *Sediment Burial:* Heavy inorganic grit and silt settle over submerged probes during low-velocity periods, locking the diaphragm in place.
3. *Vented Cable Moisture Failure:* The atmospheric reference capillary tube easily ingresses humid sewer air if desiccant cartridges are not replaced monthly, leading to internal electrical short circuits and irreversible sensor death.

2.4 Short-Range Optical Time-of-Flight (ToF) and LiDAR

Optical distance sensors transmit pulsed near-infrared laser light (typically $\lambda = 850$ nm or 940 nm) and measure photon reflection ToF via single-photon avalanche diode (SPAD) arrays [23]. In prototype testing by Utepov et al. [23], low-cost micro-LiDAR and ToF sensors demonstrated high millimeter-level accuracy under dry, clean laboratory conditions. However, when installed in live sewer manholes, optical beams suffer severe Rayleigh and Mie scattering caused by ambient water vapor, fog, and suspended aerosol droplets. Furthermore, splashing sewage coats optical protective lenses with opaque grease films within days, rendering un-wiped optical distance sensors unviable for raw sewage monitoring [23].

3 Discrete Binary Surcharge and Pulse/Float Sensors

While continuous level instruments deliver rich hydrographs, their capital, maintenance, and power requirements preclude utility-wide saturation across every sewer manhole. To achieve cost-effective spatial visibility, utilities employ discrete binary sentinels designed to signal surcharge thresholds.

Table 1: Detailed Operational Comparison: Continuous Level Sensors vs. Discrete Pulse/Float Sentinels.

Functional sion	Dimen- sion	Continuous Level Sensors (Radar / Ultra- sonic)	Discrete Pulse / Float Sentinels (Tilt / Float Switch)
Output Signal		Continuous depth time series $h(t) \in [0, D_{\text{pipe}}]$ (m)	Single-bit event alert $y(t) \in \{0, 1\}$
Dynamic Tracking	Trend	Captures diurnal flow hydrographs, dry-weather baseline, and wet-weather attenuation	Zero trend visibility; blind to all flow fluctuations below trigger height h_{alert}
Rate-of-Rise Calculation		Direct real-time calculation of $\frac{dh}{dt}$ to predict time-to-spill T_{spill}	Incapable of calculating rate-of-rise; alert is instantaneous at trigger
Power Demand		Active measurement chirp $\approx 20\text{--}100$ mW; requires strict duty-cycling	Zero quiescent power; interrupt-driven microcontroller wake-up
Battery Autonomy		2–5 years on primary Li-SOCl_2 cells (5-minute duty cycle)	5–10+ years on standard primary cells
Capital (CAPEX)	Cost	High (\$800 – \$2,500 AUD per node installed)	Ultra-Low (\$100 – \$350 AUD per node installed)
Primary Modes	Failure	False echoes, acoustic refraction, condensation dead bands	Ragging entanglement around cable; grease ball sticking
Utility Role		High-priority trunk calibration, pump stations, outfalls	Dense spatial sentinels across upstream reticulation branches

3.1 Operating Principles and Transducer Typology

Discrete level sensors operate as single-bit state switches, outputting a binary indicator $y(t) \in \{0, 1\}$:

$$y(t) = \begin{cases} 1 & \text{if } h(t) \geq h_{\text{alert}} \quad (\text{Surcharge / High-Level Trigger}) \\ 0 & \text{if } h(t) < h_{\text{alert}} \quad (\text{Normal Dry-Weather Flow}) \end{cases} \quad (5)$$

3.1.1 Mechanical Float and Tilt Switches

Mechanical switches comprise a hollow, chemically resistant polypropylene or PVDF casing suspended at a fixed elevation within the manhole chimney or overflow weir crest.

- **Reed Switch Mechanism:** A buoyant float moves along a central guide tube containing a hermetically sealed magnetic reed switch. Rising water lifts the float, aligning an internal permanent magnet with the reed switch contacts to close the circuit.
- **Mercury-Free Rolling-Ball Tilt Switch:** A weighted, pear-shaped float hangs vertically in air. When rising sewage immerses the float, buoyancy tilts the housing past a critical angle ($> 15^\circ$), causing an internal steel ball to roll across gold-plated microswitch terminals.
- **Zero Quiescent Energy Consumption:** Because dry contacts draw strictly zero electrical current during normal dry conditions, connected wireless microcontrollers remain in deep sleep modes, waking only via external hardware pin interrupts upon contact closure. This enables operational lifetimes of 5 to 10+ years on standard internal lithium cells.

3.1.2 Conductivity Pulse Electrodes

Conductivity pulse probes utilize two or more exposed metallic electrodes (titanium, Hastelloy, or 316 stainless steel) suspended at the trigger elevation. When wastewater bridges the electrodes, the high electrical conductivity of raw sewage ($\sigma \approx 500\text{--}2000 \mu\text{S/cm}$) drops circuit impedance from $> 1 \text{ M}\Omega$ to $< 2 \text{ k}\Omega$, triggering an operational amplifier comparator circuit. While conductivity electrodes contain no moving parts to mechanically jam, they are vulnerable to insulating FOG grease films that prevent electrical conduction unless periodic high-voltage capacitive discharge pulses are applied to clear grease coatings.

4 Open-Channel Flow and Velocity Sensing

Quantifying true volumetric discharge ($Q = A(h) \cdot \bar{v}$) in gravity sewers is critical for capacity assessment and infiltration quantification. However, open-channel velocity measurement is substantially more challenging than level sensing [1, 3].

4.1 Submerged Area-Velocity (AV) Ultrasonic Doppler Probes

Submerged AV probes combine a piezoresistive depth sensor with continuous-wave or pulsed acoustic Doppler transceivers mounted at the pipe invert.

- **Acoustic Doppler Shift:** The sensor emits ultrasonic beams ($f_0 \approx 500 \text{ kHz} - 1 \text{ MHz}$) into the upstream fluid. Suspended particulate matter and micro-bubbles scatter the sound back to a receiving crystal with a Doppler frequency shift Δf :

$$v_{\text{particle}} = \frac{c_{\text{fluid}} \cdot \Delta f}{2f_0 \cos \theta} \quad (6)$$

where θ is the acoustic beam inclination angle and $c_{\text{fluid}} \approx 1480 \text{ m/s}$ is sound speed in sewage.

- **Pulsed Doppler Velocity Profiling:** Advanced Acoustic Doppler Velocity Profilers (ADVP) gate returning acoustic echoes in discrete time windows, measuring velocity at multiple vertical layers across the conduit cross-section to calculate true mean velocity $\bar{v}$.
- **Field Failure Vulnerabilities:** Submerged AV probes are highly susceptible to ragging (sanitary wipes wrapping around the probe wedge), silt burial during low nocturnal velocities, and signal dropout in clear, low-turbidity waters [1].

4.2 Non-Contact Dual-Radar (Level + Surface Velocity)

To eliminate probe submergence and ragging, contactless flow monitoring stations combine an FMCW level radar with a 24 GHz planar microwave Doppler radar angled at the open-channel water surface ($\theta \approx 45^\circ$) [1, 3].

- **Surface-to-Mean Velocity Indexing:** Microwave Doppler measures surface velocity v_s . Cross-sectional mean velocity $\bar{v}$ is estimated via empirical hydraulic conversion factors:

$$\bar{v} = k \cdot v_s \quad (\text{where } k \approx 0.85 \text{ to } 0.90) \quad (7)$$

- **Hydrodynamic Breakdown Conditions:** Field investigations by Alshami et al. [1] and Blumensaat et al. [3] demonstrate that the index factor k is not constant. Under backwater conditions, hydraulic jumps, standing surface waves, or heavy grease scum crusts, the surface velocity decouples from underlying fluid conveyance, causing flow rate errors exceeding 30–50%.

4.3 The Level-Derived Flow Fallacy: Manning’s Equation Failure

A pervasive shortcut in sewer operations is deriving discharge Q purely from continuous level $h(t)$ using Manning’s open-channel formula:

$$Q = \frac{1}{n} A(h) R_h(h)^{2/3} S_0^{1/2} \quad (8)$$

where n is Manning’s roughness coefficient, $A(h)$ is wetted cross-sectional area, $R_h(h)$ is hydraulic radius, and S_0 is pipe bed slope.

Why Level-to-Flow Conversions Fail in Practice: Manning’s equation strictly assumes ****steady, uniform flow**** where the hydraulic friction slope S_f equals the physical bed slope S_0 . In real sewer networks, conditions are rarely uniform:

1. *Downstream Blockages and Chokes:* When a partial blockage occurs downstream, water level rises ($h \uparrow$) while velocity drops ($v \downarrow$). Manning’s equation interprets the rising depth as an increase in flow ($Q \uparrow$), reporting an illusory surge in discharge when the pipe is actually choking [1].
2. *Backwater and Tidal Inundation:* Upstream propagation of backwater from pump station sumps or high river levels reduces hydraulic gradient ($S_f \ll S_0$), producing discharge overestimations up to 200%.
3. *Variable Roughness:* Biofilm slime and sediment beds alter the effective roughness n seasonally ($n = 0.011 - 0.018$). True flow measurement in non-uniform reaches requires simultaneous measurement of depth and velocity.

5 Online Water Quality and Chemical Sensing

5.1 Basic Physicochemical Probes: EC, pH, ORP, and Turbidity

- **Electrical Conductivity (EC / σ):** Toroidal inductive (electrodeless) conductivity probes utilize two encapsulated wire-wound toroids inside a polymer housing. An alternating primary magnetic field induces an electric current in the surrounding fluid, which in turn induces a secondary signal in the receiving toroid. Because the coils are completely sealed in chemically inert PVDF or PEEK, inductive probes are immune to electrode fouling, scaling, and polarization [14]. Baseline EC ($\approx 800\text{--}1500\ \mu\text{S}/\text{cm}$) drops sharply ($< 200\ \mu\text{S}/\text{cm}$) during wet-weather I&I rainwater dilution events, while sudden spikes ($> 4000\ \mu\text{S}/\text{cm}$) pinpoint industrial brine dumping or coastal saltwater intrusion [20].
- **pH and Oxidation-Reduction Potential (ORP):** Measured potentiometrically via a glass-bulb hydrogen-ion-selective membrane paired with an Ag/AgCl reference electrode. In raw sewage, reference junctions experience rapid poisoning when dissolved sulfide (S^{2-}) reacts with silver chloride to precipitate insoluble silver sulfide (Ag_2S), causing severe potential drift ($> 50\ \text{mV}/\text{week}$) [11]. Solid-state polymer electrolyte junctions with double-junction barriers extend maintenance intervals from weekly to monthly. ORP values below -150 to $-300\ \text{mV}$ serve as an operational sentinel for anaerobic septic conditions favorable to sulfide production [5].
- **Optical Luminescent Dissolved Oxygen (LDO):** Operates via fluorescence/phosphorescence phase-shift quenching. Collisions with dissolved molecular oxygen (O_2) non-radiatively quench ruthenium or platinum organic complexes immobilized in an oxygen-permeable sensing cap. In comparative trials conducted by Helm et al. [9], optical luminescent sensors demonstrated complete immunity to H_2S poisoning, zero oxygen consumption (operating in static zero-flow manholes), and recalibration intervals of 6–12 months, outperforming amperometric Clark polarographic cells.
- **Turbidity and Total Suspended Solids (TSS):** Quantified optically via nephelometric light scattering ($\lambda = 860\ \text{nm}$, ISO 7027). Scattered light at 90° or 180° is measured by silicon photodiodes. Motorized wipers are mandatory in raw sewage; un-wiped optical faces become occluded by bacterial biofilms and grease films within 24–48 hours [10, 11].
- **Nutrient Ion-Selective Electrodes (ISEs):** PVC membranes doped with nonactin measure ammonium (NH_4^+). Potassium ions (K^+) present severe cross-interference due to similar ionic radii ($r_{\text{K}^+} \approx 133\ \text{pm}$ vs. $r_{\text{NH}_4^+} \approx 143\ \text{pm}$), requiring secondary potassium ISEs for mathematical cross-compensation [22].

5.2 Advanced Optical Spectroscopy for Organic Load Monitoring

Traditional laboratory measurements of organic load (BOD_5 , COD, and TOC) require slow 5-day incubation or 2-hour chemical digestion. In-situ optical spectroscopy provides instantaneous, reagent-free surrogates [4, 10, 14].

- **UV-Vis Spectrophotometry (200–750 nm):** In-situ spectrometer probes measure continuous multi-wavelength absorbance across a narrow path length (1–2 mm in raw sewage). Dissolved organics with conjugated double bonds absorb strongly in the UV range ($\lambda = 254\ \text{nm}$), nitrate absorbs at deep UV ($\lambda = 210\text{--}220\ \text{nm}$), and particulate solids cause visible scattering (400–750 nm). Chemometric deconvolution (PLSR) converts raw absorption spectra into simultaneous continuous proxies for COD, TOC, NO_3^- , and TSS [10].
- **2D-Fluorescence (EEM) Spectroscopy:** Measures emitted fluorescence across excitation-emission matrix coordinates. Protein-like fluorophores (**Tryptophan-like Peak T**, $\lambda_{\text{ex}}/\lambda_{\text{em}} \approx 280/350\ \text{nm}$) correlate strongly ($R^2 > 0.85\text{--}0.95$) with readily biodegradable biochemical oxygen demand (BOD_5) [4]. Peak T provides an instantaneous (< 5 second) optical proxy for BOD_5 , eliminating the 5-day analytical delay for real-time treatment plant influent monitoring.

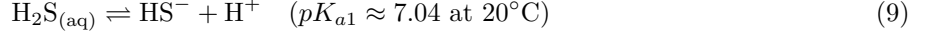
5.3 Biological and Bioelectrochemical Sensors

Microbial Fuel Cell (MFC) biosensors utilize electrochemically active bacteria (anodophilic exoelectrogens, e.g., *Geobacter sulfurreducens*) immobilized as an anaerobic biofilm on a conductive carbon anode [21]. Microorganisms metabolize dissolved organic matter in the wastewater, transferring metabolic electrons directly to the anode to generate a measurable electric current $I_{\text{cell}} \propto \text{BOD}$. While directly quantifying biologically assimilable organics, MFC biosensors experience dynamic saturation at high BOD ($> 300\ \text{mg}/\text{L}$) and response time lags (10–30 minutes) governed by microbial metabolic kinetics [21].

6 Hydrogen Sulfide (H_2S) and Biogenic Corrosion Sensing

Biogenic sulfuric acid corrosion in concrete sewer crowns is driven by hydrogen sulfide (H_2S) emissions [15, 16].

- **Sulfide Generation and Aqueous Equilibrium:** Under anaerobic conditions in submerged sewer biofilms and force mains, sulfate-reducing bacteria (SRB) reduce sulfate to dissolved sulfide (S^{2-} , HS^-). Liquid-phase speciation follows aqueous pH equilibrium:



At typical wastewater pH ≈ 6.8 – 7.2 , approximately 40%–60% of total dissolved sulfide exists as molecular $H_2S_{(aq)}$, which strips across the liquid-gas interface into the sewer headspace:

$$r_{\text{emission}} = K_L a \cdot \left([H_2S_{(aq)}] - \frac{C_{\text{gas}}}{H} \right) \quad (10)$$

where $K_L a$ is the mass transfer coefficient (heavily elevated by hydraulic drop structures and turbulent bends) and H is Henry’s law constant. Headspace H_2S is oxidized by aerobic sulfur-oxidizing bacteria on concrete walls to sulfuric acid (H_2SO_4), degrading concrete crowns at rates up to 5–10 mm/year [16].

- **Headspace Gas Sensors:** Amperometric electrochemical gas cells provide sub-ppm sensitivity (0.1–500 ppm) at micro-watt power. However, continuous condensing humidity (100% RH) blinds gas membranes with liquid water droplets, while exposure to severe sulfide spikes (> 500 ppm) poisons the electrolyte, causing rapid baseline drift [15].
- **Liquid-Phase In-Situ UV Bisulfide Probes:** Submerged UV spectrometer probes measure bisulfide ion absorption in the deep UV spectrum ($\lambda \approx 230$ nm). Despot et al. [7] demonstrated that submerged UV probes with automated wipers operate reliably in raw sewers for over six months, providing direct feedforward chemical dosing control (e.g., iron salts or oxygen injection) that reduced chemical consumption by 35% compared to retrospective gas-phase monitoring.

7 Pipe Condition, Blockage, and Structural Screening

7.1 Acoustic Pulse Reflectometry

Acoustic pulse reflectometry is a rapid, non-invasive acoustic screening technique for sewer conduits [2, 19, 24]. An acoustic transceiver mounted in a manhole transmits a high-energy acoustic pulse (100 Hz–5 kHz) down the pipe headspace. When the sound wave encounters a change in cross-sectional acoustic impedance—such as a silt deposit, FOG accumulation, tree root intrusion, or lateral fracture—a portion of the acoustic energy reflects back to a microphone array [2]:

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{A_1 - A_2}{A_1 + A_2} \quad (11)$$

where $Z = \frac{\rho c}{A}$ is acoustic impedance, and A_1, A_2 are cross-sectional headspace areas before and at the obstruction. Bin Ali et al. [2] and See et al. [19] demonstrated that acoustic pulse reflectometry can screen up to 100 meters of pipe in under 30 seconds from a single manhole, identifying $> 10\%$ cross-sectional blockages without CCTV crawler deployment.

7.2 Distributed Fiber-Optic Sensing (DTS/DAS)

Fiber-optic sensing cables installed along the crown or invert of sewer mains turn the optical fiber into a continuous sensor array [17]:

- **Distributed Temperature Sensing (DTS):** Uses Raman backscattering optical time-domain reflectometry (OTDR) to measure temperature profiles every 1 meter along kilometers of pipe. Localized thermal anomalies (0.5°C shifts) pinpoint lateral illicit connections and cold groundwater infiltration plumes [17].
- **Distributed Acoustic Sensing (DAS):** Uses Rayleigh coherent optical time-domain reflectometry (ϕ -OTDR) to detect acoustic vibrations induced by bedload sediment movement, gas bubbling, and structural cracking.

7.3 CCTV Inspection and Automated Computer Vision

Closed-Circuit Television (CCTV) robotic crawlers equipped with high-definition cameras and structured laser profilers capture internal pipe imagery. Convolutional neural networks (CNNs) and transformer models automatically segment cracks, displaced joints, root intrusions, and FOG accumulation from video logs [13]. However, CCTV remains a periodic, episodic inspection method (conducted every 5–10 years) and cannot provide real-time hydraulic alert capability.

8 Environmental Degradation, Telemetry, and Maintenance

8.1 Biofouling and Grease Encrustation Dynamics

In raw sewage, bacterial colonization forms thick biofilms within 48 hours. FOG molecules precipitate onto surfaces, forming sticky grease crusts.

- *Non-Contact Sensors:* FMCW radar and acoustic sensors mounted in the manhole chimney are physically decoupled from the liquid, preventing direct biofouling. However, splashing during peak storms can deposit grease on transducer lenses, necessitating annual visual checks [8].
- *Submerged Contact Probes:* Hydrostatic, AV Doppler, and optical probes require automated mechanical wipers, ultrasonic self-cleaning horns, or pressurized air-purges to prevent fatal measurement drift [11, 14].

8.2 Subterranean Telemetry and Radio Propagation

Moving data from underground sewer manholes to cloud platforms is constrained by radio frequency (RF) physics [8, 12]:

- **LoRaWAN (868 / 915 MHz):** Long-range, low-power spread spectrum modulation. Demonstrates high penetration capabilities through concrete manhole shafts, achieving successful transmission over 3–8 km line-of-sight from manhole chimneys to elevated gateway towers [8].
- **Cellular IoT (NB-IoT / LTE-M):** Operates on licensed cellular spectrum (700–900 MHz) with narrow carrier bandwidths, providing high link margins (+20 dB over standard GSM).
- **Manhole Cover Attenuation:** Solid cast-iron covers reflect > 99% of RF energy. Practical mitigation involves mounting antennas directly beneath composite non-metallic covers, using external low-profile puck antennas embedded in road asphalt, or suspending dipole antennas in the manhole chimney neck [8].

8.3 Battery Energy Budgeting and Duty Cycling

Long-term deployment mandates 3–7 years of maintenance-free operation on internal $Li - SOCl_2$ battery packs (e.g., 19 Ah at 3.6 V). The total energy consumption E_{total} is governed by:

$$E_{\text{total}} = N_{\text{sample}} (E_{\text{warmup}} + E_{\text{measure}}) + N_{\text{tx}} (E_{\text{tx}}(P_{\text{tx}}, \text{SF})) + E_{\text{sleep}} \quad (12)$$

where SF is the spreading factor and P_{tx} is transmission power. Adaptive sampling algorithms—keeping transmission intervals at 30–60 minutes during stable dry-weather flow and dynamically accelerating to 1-minute intervals upon detecting a level rate-of-rise $\frac{dh}{dt} > \theta_{\text{rise}}$ —extend battery lifespan while preserving high temporal resolution during storm events.

9 Comprehensive Comparative Synthesis

Tables 2 and 3 present an exhaustive comparative synthesis across all physical, hydraulic, chemical, optical, and gas sensing modalities evaluated in this review.

10 Conclusion

This comprehensive review has evaluated the full spectrum of physical, hydraulic, chemical, optical, and gas-phase sensing technologies developed for municipal wastewater collection networks. The core conclusions are:

1. **Continuous Level Sensing Dominance:** 77–81 GHz FMCW millimeter-wave radar provides the most resilient non-contact depth measurement, completely avoiding the acoustic refraction, condensation blinding, and foam dissipation that impair ultrasonic transducers in deep, humid manholes.

Table 2: Comparison of Physical, Hydraulic, Level, Flow, and Structural Sensors in Wastewater Systems.

Sensing Modality	Physical Principle	Parameter	Parameter	Power Demand	Deployment	Primary Failure Modes in Wastewater	Maintenance Requirements	Cost Tier
FMCW Radar (77–81 GHz)	Microwave frequency ($f_b \propto d$)	beat	Continuous depth $h(t)$		Medium (≈ 50 – 100 mW chirp)	Spurious reflections in narrow shafts; heavy grease crust bridging	Non-contact; annual optical check; 5+ yr battery	High (\$1.2k–\$2.5k)
Ultrasonic Level (20–200 kHz)	Acoustic time ToF	transit-	Continuous depth $h(t)$		Low–Med (≈ 20 – 50 mW)	Thermal stratification refraction (5–12% error); face condensation blinding; foam absorption	Bi-monthly lens wipe; temperature calibration	Medium (\$400–\$900)
Hydrostatic Pressure Probe	Submerged piezoresistive diaphragm		Water column head $h(t)$		Very Low (≈ 5 – 15 mW)	Diaphragm FOG encrustation; sediment burial; capillary vent moisture ingress	Monthly physical cleaning; desiccant replacement	Low–Med (\$300–\$700)
Mechanical Float / Tilt Switch	Microswitch / rolling ball circuit closure	ball	Binary charge $y(t)$	sur-alert	Ultra-Low (Zero quiescent sleep)	Ragging entanglement around cable; grease ball sticking	Semi-annual mechanical free-movement test	Very Low (\$100–\$250)
Submerged AV Doppler Flow	Acoustic Doppler backscatter + depth		Velocity $\bar{v}$ + depth h		Medium (≈ 80 – 150 mW)	Silt burial; ragging entanglement; acoustic dropout in clear flow	Monthly confined entry; rag removal	High (\$2.5k–\$5k)
Contactless Dual-Radar Flow	FMCW + planar microwave Doppler	level	Surface velocity v_s + depth h		Med–High (≈ 100 – 300 mW)	Standing waves; k -factor breakdown under backwater	Quarterly beam alignment; velocity cal	High (\$3k–\$6k)
Acoustic Pulse Reflectometry	Acoustic waveguide impedance discontinuity		Pipe blockage, silt, cracks		Low (≈ 30 – 60 mW)	Lateral junction echo interference; high ambient acoustic noise	Non-contact; annual transceiver check	Med (\$1.5k–\$3.5k)

- Indispensability of Float/Pulse Sentinels:** Discrete mechanical float and tilt switches provide an ultra-low-power, zero-quiescent-current solution with 5–10+ year battery lifespans, enabling dense spatial sentinel coverage across upstream residential branches.
- Volumetric Flow Measurement Realities:** Manning’s equation level-to-flow conversions fail dramatically during downstream chokes and backwater. Reliable discharge quantification requires simultaneous velocity measurement via submerged AV Doppler or non-contact microwave surface Doppler.
- Rapid Optical Water Quality Surrogacy:** UV-Vis spectrophotometry and 2D-fluorescence EEM spectroscopy (targeting tryptophan-like Peak T) successfully deliver instantaneous, reagent-free proxies for COD, TOC, and 5-day BOD₅, eliminating laboratory incubation lags.
- Corrosion Mitigation:** Liquid-phase in-situ UV bisulfide (HS^-) probes enable proactive, feedforward chemical dosing control before toxic H_2S gas strips into the sewer headspace.

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Table 3: Comparison of Physicochemical, Optical, Biological, and Gas-Phase Sensors in Wastewater Systems.

Sensing Modality	Operating Principle	Target Analyte	Power Demand	Primary Failure Modes in Wastewater	Maintenance Requirements	Cost Tier
Inductive Toroidal EC	Electrodeless dual-toroid magnetic induction	Total dissolved ions (σ)	Low ($\approx 20\text{--}50$ mW)	Heavy scaling encrustation; temperature sensor failure	Monthly wipe; 5+ yr lifespan; low drift	Medium (\$600–\$1.2k)
Potentiometric pH / ORP	Glass bulb + $Ag/AgCl$ reference potential	Acidity, redox potential	Low ($\approx 10\text{--}30$ mW)	Reference junction Ag_2S sulfide poisoning; glass grease coating	Bi-weekly cleaning; monthly buffer cal	Low–Med (\$400–\$800)
Optical Luminescent DO	Phase-shift phosphorescence quenching	Dissolved oxygen (O_2)	Low ($\approx 15\text{--}40$ mW)	Biofilm growth on luminophore cap; photo-bleaching	Monthly cap wipe; annual cap replacement	Med–High (\$1.2k–\$2.5k)
UV-Vis Spectrophotometer	Multi-wavelength absorbance (200–750 nm)	COD, TOC, NO_3^- , TSS	High ($\approx 1\text{--}5$ W scan)	Optical window biofouling; matrix turbidity scattering; lamp aging	Automated wiper mandatory; monthly clean	Very High (\$8k–\$25k)
2D-Fluorescence (EEM)	Fluorophore intensity (Peak T / Peak C)	Rapid BOD ₅ surrogate	High ($\approx 2\text{--}8$ W)	Inner-filter quenching; window biofouling; matrix shifts	Frequent optical clean; site calibration	Very High (\$15k–\$40k)
MFC Biosensor (BOD)	Microbial extracellular electron transfer current	Biodegradable organic BOD	Low–Med ($\approx 10\text{--}30$ mW)	Biofilm saturation (> 300 mg/L); metabolic lag (10–30 min)	Monthly nutrient flush; active research	Med–High (\$2k–\$5k)
Headspace Gas H_2S	Amperometric oxidation at gas electrode	Gas-phase H_2S (0.1–500 ppm)	Very Low ($\approx 5\text{--}20$ mW)	Electrolyte dry-out; condensation blinding; high-ppm poisoning	Quarterly recalibration; annual replacement	Medium (\$600–\$1.5k)
Liquid Bisulfide Probe	In-situ UV absorbance at $\lambda \approx 230$ nm	Dissolved HS^- for chemical dosing	Med–High ($\approx 1\text{--}3$ W)	Severe window biofouling; wastewater color interference	Automated wiper mandatory; monthly check	High (\$5k–\$12k)

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